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Polynyas and Leads: An Overview of Physical Processes and Environment

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Polynyas and leads are openings in pack ice due to divergences in ice drift and to local melting. They are the vents and windows to the polar oceans. In winter they are a major source of brine during freezing and a locus for gas exchange. Large sensible heat fluxes, together with evaporation and longwave radiation from a very small percentage of open water and thin ice, dominate regional heat budgets. In summer, solar radiation is absorbed by open water but is reflected from snow-covered pack ice. Experiments and models describing these processes are reviewed.

1. INTRODUCTION

At predictable, recurrent locations throughout the polar regions, there are oceanic areas which remain either partially or totally ice free at times and under climatological conditions where we would expect the waters to be ice covered. These areas are called polynyas [*World Meteorological Organization* (WMO), 1970]. Polynyas appear in winter, when air temperatures are well below the freezing point for sea water, and are bordered by water which is ice covered. Polynyas are typically rectangular or elliptical and occur quasi-continuously in the same region. They range in size from a few hundred meters to hundreds of kilometers. In certain areas, such as the portion of the Beaufort Sea west of Banks Island, polynyas may gradate into preferred areas for lead formation [*Markham*, 1981].

Polynyas are of interest for a number of reasons encompassing both physics and biology. For example, many polynyas, particularly those bordering Arctic and Antarctic lee coastlines during winter, are sites for active brine formation. This brine may affect the local water density structure and current field and may also influence larger-scale water mass modification as it is mixed and advected beyond the polynya region. The relatively regular occurrence of polynyas makes them important as biological habitats [*Stirling*, 1980; *Massom*, 1988]. The associated open water can contribute in spring to a localized plankton bloom which can "seed" downcurrent areas. Large mammals use them as feeding grounds, suggesting that rich benthic activity takes place as well. Inuit people have used polynyas as hunting grounds for the past 3000 years [*Schledermann*, 1980].

Polynyas can form by two mechanisms. First, ice may form within the region and be continually removed by winds, currents, or both. The heat required to balance loss to the atmosphere and hence to maintain the open water is provided by the latent heat of fusion of the ice which continually forms. In the second mechanism, oceanic heat may enter the region in sufficient quantities to locally prevent ice formation. The first mechanism creates polynyas referred to as "latent heat" polynyas, and the second mechanism creates "sensible heat" polynyas. These two mechanisms are not mutually exclusive, and both contribute in many cases to the maintenance of polynyas.

Leads, in contrast to polynyas, are openings in pack ice which form as a result of local divergences in its drift. They occur at weak points in the ice and characteristically take the form of long, narrow channels which are meters to hundreds of meters wide and kilometers to tens of kilometers long. They can make it possible for vessels to navigate in seasons and in areas where only heavy icebreakers could otherwise penetrate, and they serve also at times as corridors for migration of marine mammals. Ice divergence, the process primarily responsible for formation of latent heat polynyas, is also the process responsible for lead formation. In this respect, then, latent heat polynyas and leads are similar.

Leads have no fixed locations, even though their occurrence may be predictable by region and season [e.g., *Marko and Thomson*, 1975, 1977]. They are more prevalent in areas of thinner ice and in the marginal ice zones. Even in the central polar pack ice in winter, leads typically cover at least 1% of the area. Typical ice cover is shown in ice atlases; for Alaskan waters, see *LaBelle et al.* [1983], and for Canadian and western Greenland waters, see *Markham* [1981] and *Ito* [1982*b*]. *Wadhams* [1981] reviewed the ice cover of Greenland and Norwegian seas. *Wadhams et al.* [1985] reported that the average distance between leads varies from 5 km in the marginal ice zone to as much as 275 km in the central Arctic pack ice. *Thorndike et al.* [1975] have modeled the ice thickness distribution of the Beaufort Sea over a 2-year period (May 1962 to April 1964) using data from U. S. and Soviet drifting stations, and estimate about 1% of 1- to 10-cm ice cover during the winter months. *McLaren* [1988] re-

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ported 2.6% of open water and new ice (<30 cm) from a submarine traverse under the North Pole in summer.

A newly formed lead in the winter pack ice can freeze over within less than a day [*Lebedev*, 1968; *Bauer and Martin*, 1983] or can close up without warning in response to converging ice drift. At other times the processes which initially open up a lead can continue to keep it open for weeks. The thin ice of a frozen-over lead remains thinner and weaker than the surrounding pack ice, even though it grows faster. Converging ice drift forces can cause a refrozen lead to fail in compression, piling up ice blocks which freeze together as an ice ridge [*Parmerter and Coon*, 1972].

Heat exchange at the open water surface of a lead or polynya is 2 orders of magnitude greater than that through surrounding snow covered pack ice, so even a small percentage of these features and of thin ice can dominate regional heat budgets in winter [*Maykut*, 1978; *Ledley*, 1988]. Release of heat from the ocean is one of the principal reasons why north polar winters are less frigid than those of the Antarctic continent, even near sea level. In the spring and summer, open water surfaces absorb over 90% of incoming solar radiation while snow-covered ice has a typical albedo of about 0.8. Leads and polynyas then provide centers of initial warming prior to the spring breakup of ice cover [*Hall and Rothrock*, 1987]. The influence of these features on the heat budget will be a central theme of this review.

The physics of Arctic pack ice, leads and polynyas are reviewed by *Pritchard* [1980] and by *Untersteiner* [1986]. *Jacobs* [1985] gives an overview of work in the Antarctic. This review is limited mainly to the studies of leads and polynyas reported in the Western literature, and does not cover the very substantial contributions of Soviet scientists [e.g., *Makshtas*, 1984].

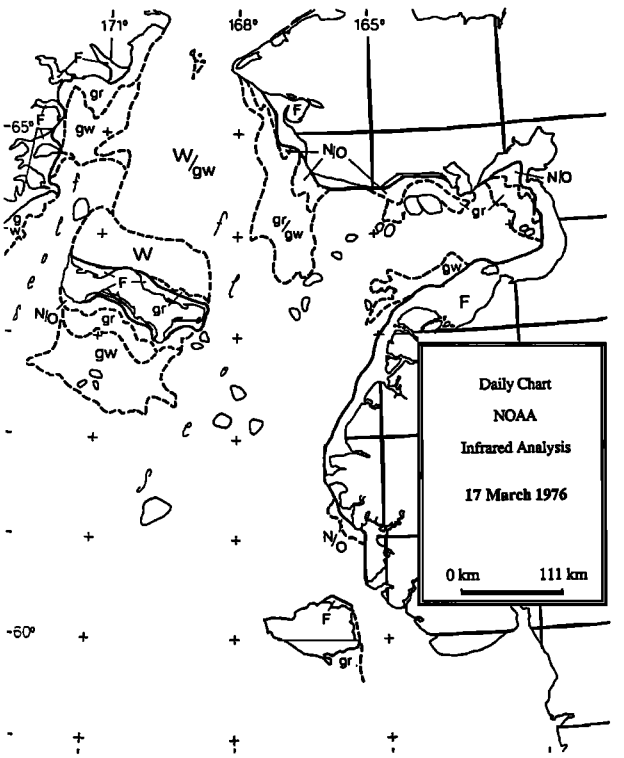

Fig. 1. Coastal polynyas in the northern Bering Sea, analyzed from satellite imagery. The letters indicate ice thickness categories: N, new; O, open, gr, grey; gw, grey white; W, white; F, fast. The remaining area is first-year ice. The St. Lawrence Island polynya is seen at 170°W, 63°N [*McNutt*, 1981].

2. Latent Heat Polynyas

Latent heat polynyas form in areas in which ice is removed as quickly as it forms, exporting latent heat of fusion to surrounding areas. Drift of ice into the polynya area may be inhibited in several ways, for example, by the presence of a shoreline, by shore-fast or glacial ice, or by pack ice blocking a narrow channel. Locally formed frazil ice or ice floes drift away under the combined action of winds and currents. Well-known latent heat polynya regions occur in the Arctic off lee coastlines in the northern Bering Sea, and overlying Ob Bank just off the northeast Greenland coast. The North Water polynya in northern Baffin Bay is largely a latent heat type but may be due in part to upwelling of warm water. In the Antarctic a polynya generated in Terra Nova Bay by the interactions among wind, glacial ice, and shoreline topography has been studied [*Bromwich and Kurtz*, 1984; *Kurtz and Bromwich*, 1985]. Lee coastline polynyas have been observed along other portions of the Antarctic coast and, in particular, downcurrent or downwind from large grounded tabular icebergs.

Northern Bering Sea Polynyas

The coastal polynyas in the northern Bering Sea provide perhaps the simplest and most studied cases of latent heat polynyas. They appear to be due solely to the north-northeasterly winter winds sweeping ice away from lee coastlines as it is formed (Figure 1). Monthly mean winter winds in the northern Bering Sea are low (5–15 cm/s). Flow is northward through Anadyr and Bering straits and very small or southward for much of the winter in Shanberg Strait [*Muench et al.*, 1988]. Currents in the polynya region south of St. Lawrence Island are negligibly small through the winter. Since the shelf waters are cooled to the freezing point by midwinter, there is no oceanic heat source to prevent ice formation. Rather than being continually ice free, the northern Bering Sea polynyas (Figure 1) open up during periods of northerly to easterly winds and close when winds are southerly [*McNutt*, 1981]. Southerly winds advect existing pack ice into the region, closing the polynyas. Size is also restricted when the surface radiation balance is strongly upward (rapid cooling) during periods of clear skies and low wind speeds, so that ice forms rapidly compared to its removal by wind and nearly covers the open water area [*Pease*, 1987]. The features can be reformed rapidly once significant northerly winds occur. The mobile, fractured ice which typifies the northern Bering Sea is aerodynamically rough [see *Overland*, 1985], so that wind drag on the ice quickly becomes sufficient to move it away from the coastline. Internal compression stresses within the ice can be presumed negligible in the northern Bering under the influence of northerly winds because the Bering Sea ice pack is divergent toward the southern marginal ice zone. Once an open water surface is present following initial wind-driven ice divergence, radiation stress due to locally wind-generated surface waves can be expected to contribute to the wind forces acting on the downwind ice edge. The stress balance involving surface waves has been computed for divergent ice bands along the marginal ice zone by *Martin et*

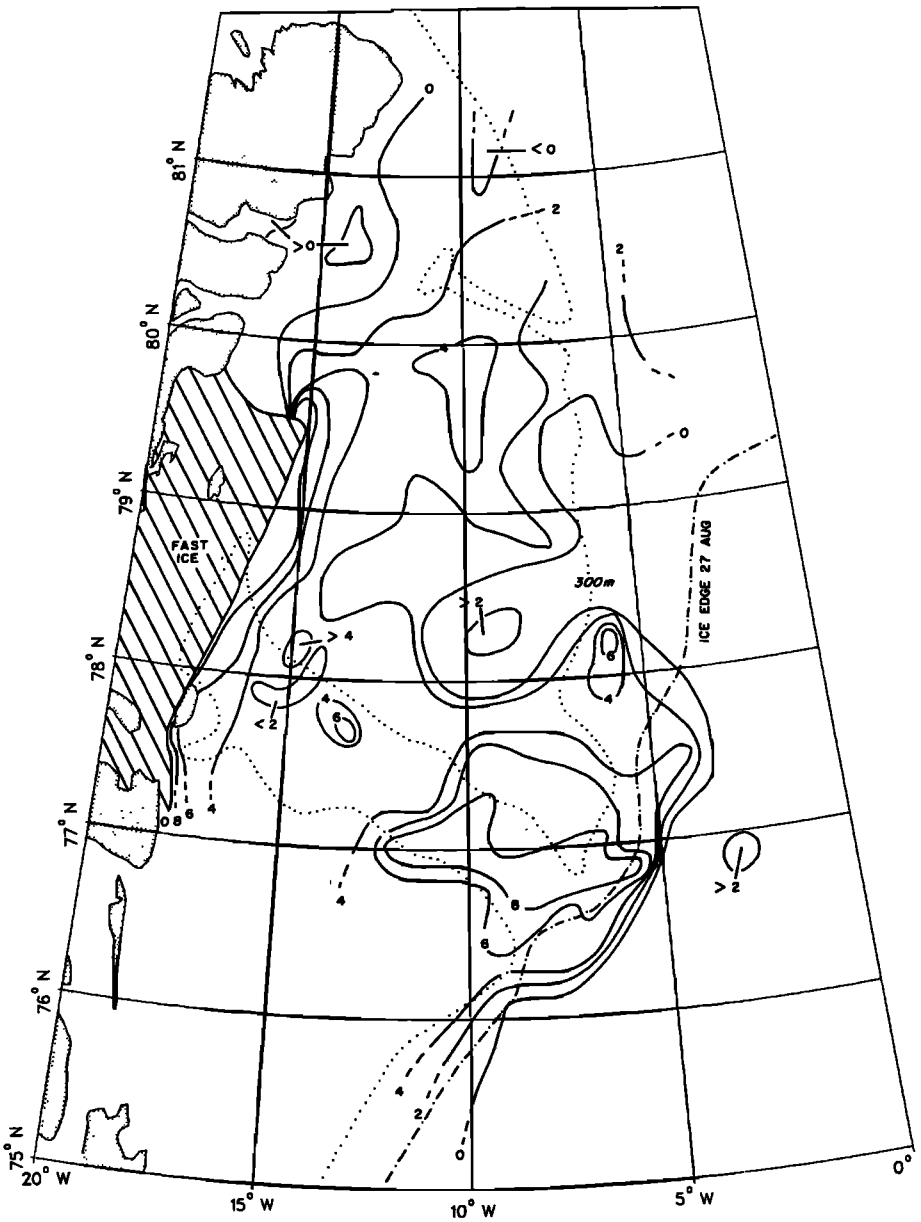

Fig. 2. Ice cover (in tenths) east of northeastern Greenland in August 1984. The Ob Bank polynya is centered at approximately 80°30″ N, 14°W. The Northeast Water is the area of low ice cover immediately off the shore or the shorefast ice, and extending approximately 77° to 81°N [*Bourke et al.*, 1987].

al. [1983], but it is not clear in the case of a polynya to what extent the presence of frazil ice would either prevent surface wave generation or damp existing waves.

Removal of ice from the polynya as it is formed allows a very large net overwinter ice formation within the polynya area. *Schumacher et al.* [1983] have estimated a net overwinter ice formation of about 5 m for the St. Lawrence Island polynya, and *Pease* [1980] calculated that the Bering Sea ice cover may be regenerated in this manner between two and six times during the ice season. The ice is driven southward by the prevailing northerly winter winds and eventually melts in the marginal ice zone. The brine rejected by this considerable ice formation here and at other locations along the lee coastlines of the northern Bering Sea strongly conditions the water on the northern central Bering Sea shelf [*Muench et al.*, 1988] and may contribute significantly to maintenance of the Arctic Ocean halocline [*Aagaard et al.*, 1981]. Ice formation events are reflected in the water column as peaks in the salinity, caused by brine rejection. Peaks in the local ocean currents can occur because of the local baroclinic field generated by elevated salinity during these events [*Schumacher et al.*, 1983].

The Northeast Water and Other East Greenland Coastal Polynyas

A polynya has been observed overlying Ob Bank just off the extreme northeastern Greenland coast [*Wadhams*, 1981; *Vinje and Finnekåsa*, 1986; *Hirche and Bohrer*, 1987] (Figure 2). We refer to this feature as the Northeast Water. This is the northernmost and largest (about 44,000 km^2) of several similar polynyas which occur along the east Greenland

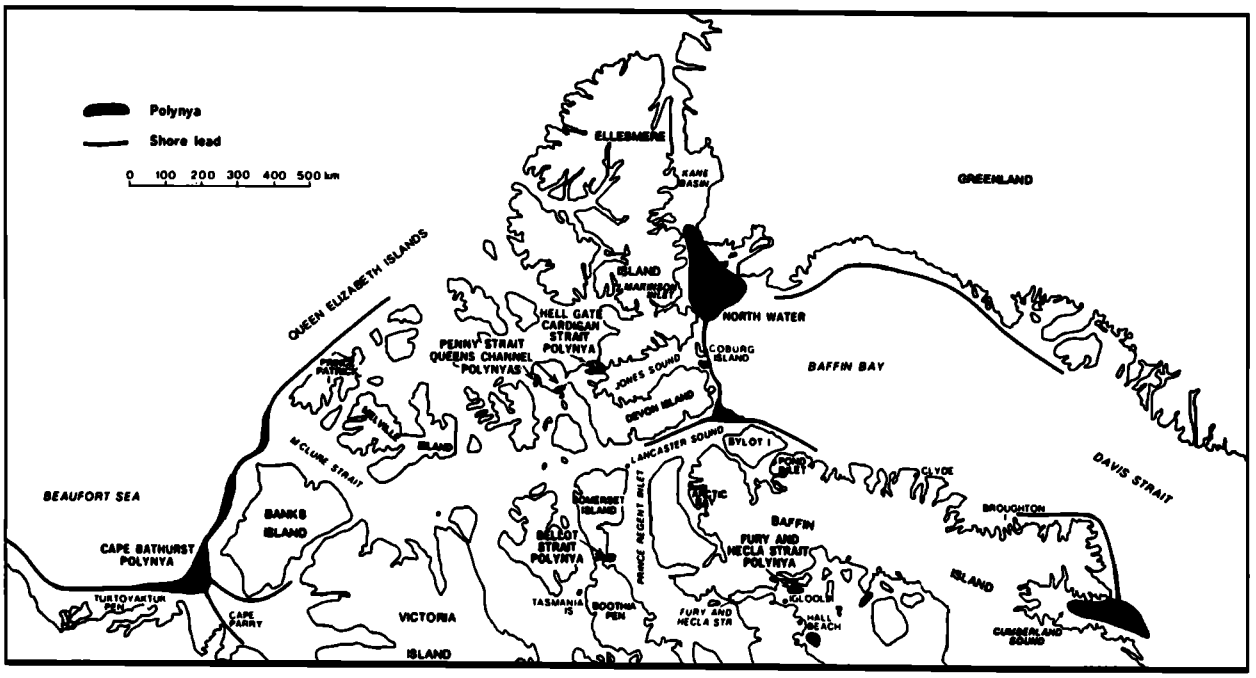

Fig. 3. Polynyas and shore leads in the Canadian Arctic [*Topham et al.*, 1983].

coast. It is characterized through winter by a thin, broken ice cover which by April or May has typically become open water. This open water generally persists through the summer until about September. On a few occasions the polynya has appeared as well in early winter (October–November).

The Northeast Water appears to be primarily a latent heat polynya. Northerly winds in winter and early spring, in conjunction with local currents associated with an anticyclonic gyre over the shelf to the south and concentrated SE flow along the shelf break (see Figure 2), remove ice from the coastal region. The bank underlying the polynya is sufficiently shallow that deep-draft ice in the form of tabular icebergs or ice islands which have broken loose from glaciers in the Canadian archipelago can become grounded, preventing ice from advecting into the area from the north. When this happens, it significantly expands the size and duration of the polynya. Such a condition was observed to persist from mid-1981 until late 1985 and might be quite common given the large number of icebergs in the region.

It is possible, though not demonstrated, that oceanic heat may at times contribute to formation of the Northeast Water. Recent summer oceanographic data presented by *Bourke et al.* [1987] suggest that warm water from east of the polar front is advected into the polynya region by the circulation in the deep trough south of Ob Bank. This warm water is present in summer, however, at depths of 100–200 m below the surface and underlying a strongly salinity-stratified (hence density stratified) upper layer, which would be expected to inhibit upward heat transfer. The data are inadequate to determine whether heat from this warm water layer is available at the surface at other times of the year to inhibit ice formation. Physical reasoning suggests that vertical stability of the water column would be lower in winter, upper layer salinity (and density) having been increased through ice formation. Sufficiently decreased vertical stability might allow heat to mix upward from the warm layer to the surface. Localizing shoaling of the bottom over Ob Bank might contribute to this mixing, as currents locally intensify and generate turbulence while passing over the shoal. It is possible that the mean flow interacts with shoaling bottom topography to produce localized upwelling. Passage of flow over the shoal would produce a tendency toward anticyclonic rotation, which would be expected to cause upward doming of isotherms over the bank and bring the warm Atlantic Water closer to the surface. Finally, coastal currents induced by input of low-salinity water from terrestrial melting would be expected to supply heat to the upper water column and influence the ice distribution by advection. No data are available on local coastal currents, and additional oceanographic data, in particular for winter and early spring, are needed from the region in order to investigate these possible effects.

The smaller coastal polynyas which occur farther south along the eastern Greenland coast are not well documented, although their presence is confirmed by satellite imagery. They are almost certainly latent heat types driven by strong offshore drainage winds which originate on the Greenland ice cap. These winds can be channeled through fjordal valleys so that speeds are accelerated [e.g., *Ito*, 1982*a*] and, in fact many of the more predominant East Greenland coastal polynyas occur off fjord mouths. There is no evidence for significant oceanic heat input which might inhibit ice formation.

The North Water

The North Water, which has no homogeneous, substantial ice cover during the winter and spring months, is a large region in Baffin Bay and Smith Sound situated between Greenland on the east and Ellesmere and Devon islands on the west (Figure 3). It is not defined in summer, when northern Baffin Bay is ice free. The northern boundary is a well defined structural ice arch which blocks southward flow of ice through Smith Sound [*Ito and Müller*, 1977]. Ice forms

continually within the polynya through the winter and drifts southward under the combined influence of local northerly winds and southward currents [*Dunbar*, 1969; *Muench*, 1971; *Dunbar and Dunbar*, 1972; *Muench and Sadler*, 1973]. The wind effect is accentuated as winds attain high speeds by channeling through steep sided valleys [*Ito*, 1982*a*, 1985]. It can also be expected, on the basis of the work of *Martin et al.* [1983], that radiation stress from surface waves contributes to sweeping ice from the polynya. This effect might be relatively more significant in the North Water than in smaller polynyas. Downwind fetch in the North Water can exceed 100 km at times, which would allow for well-developed local seas and consequently large associated stress. In contrast to the well-defined arch in Smith Sound, the southern extent of the polynya is poorly defined, since it is the recipient of the drifting pack which forms within. In spring the polynya expands southward as the ice melts, finally joining the northward expansion of open water associated with the West Greenland current. Like other coastal polynyas which are highly dependent upon synoptic meteorological processes, the North Water is not continually present but occurs in pulsewise fashion.

Early work on the North Water, based upon summertime oceanographic data, suggested that there was no oceanic heat source adequate to prevent ice formation in winter [*Muench*, 1971]. Adequate heat appears to be advected into northern Baffin Bay by the warm West Greenland Current to significantly delay ice formation in the fall, though available observations are inadequate to quantitatively determine the northward heat transport. Available heat is contained, in summer, in a layer of warm Baffin Bay Atlantic Water at depths greater than 100 m, overlain by a stable, salinity-stratified upper layer which isolates it from the surface.

As part of the North Water Project (1972–1981), mesoscale field and modeling studies of the ice, ocean and atmosphere in this region were carried out by a Canadian-Swiss team [*Crawford and Parkinson*, 1981; *Müller et al.*, 1977] in 1978–1981. Contrary to previous expectations [*Müller et al.*, 1977], this region does not remain fully ice free in winter and typically has some 90% ice cover (Figure 4). This includes a substantial portion of first-year ice floes over 30 cm thick, together with a variety of thinner ice types [*Steffen*, 1986]. As such, the North Water may not meet the strict definition of a polynya. Nevertheless, the relatively thin and loose ice, as compared to other oceanic areas at comparable latitudes, results in an area which remains generally navigable by ice-reinforced vessels and which has an anomalously large surface heat exchange. Three smaller polynyas have been identified within the North Water region: the Smith Sound, Barrow Strait, and Lady Ann Strait polynyas. Each of these lies south or east of narrow passages which constrict the inflow of pack ice, while locally formed ice and that which does come through the narrow passages is free to drift under the influence of winds and currents to the southern edge of the North Water (approximately 75°N).

There are no winter oceanographic data, beyond the surface skin temperatures described below, but the general drift is believed to be toward the southeast. The mean winter wind on the western side of Smith Sound was reported by *Steffen and Ohmura* [1985] to be 9 m/s, and gales (>20 m/s and up to 40 m/s) from the NNW occurred on the average every tenth day with a mean duration of 19 hours.

In addition to removal of latent heat by the formation and drift of ice, upwelling of warmer water has been hypothesized to be a major source of heat. By airborne remote sensing, *Steffen* [1985*b*] found cells of surface water, with temperatures ranging from −1.7 to −0.1°C, covering up to 5% of the path surveyed. These warm water cells were most prevalent in eastern Smith Sound, where Coriolis deflection of ice and surface water away from the shore under the influence of northerly winds can be expected to result in coastal upwelling of warmer underlying water. *Steffen* [1985*b*] estimated that such coastal upwelling might amount to a heat supply of some 3×10^{12} W over the entire area, as compared with a surface heat loss of 7×10^{12} W. *Steffen and Ohmura* [1985] speculate that only 20% of the oceanic heat flux is due to the freezing of sea water and 80% is due to supply of heat from the warmer sea water. It is clear that both mechanisms are important but that winter studies of water masses, currents, and ice drift will be needed to resolve their relative contributions.

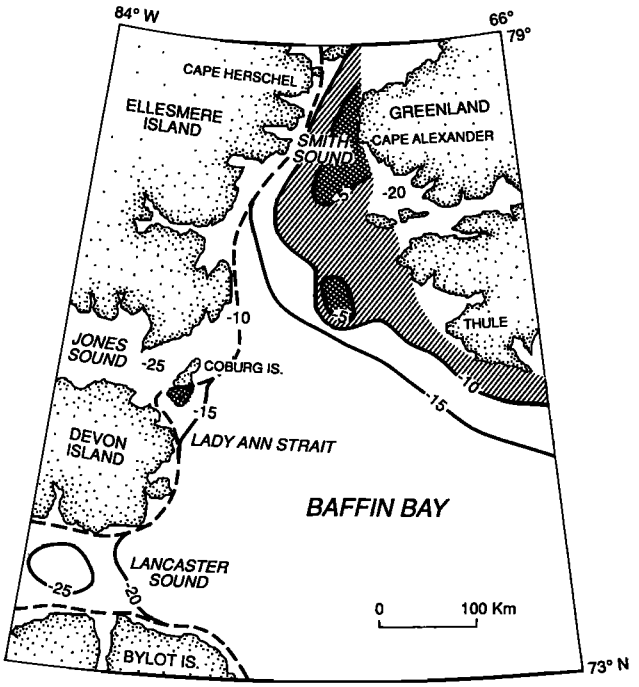

Fig. 4. Effective mean surface temperature contours (°C) of the North Water region on January 22, 1981. Cross-hatched areas, $T > -5$°C; hatched areas, $-10 < T < -5$°C (adapted from *Steffen and Ohmura* [1985]).

The Terra Nova Polynya

The Terra Nova Bay Polynya in the Antarctic is likewise maintained by strong offshore winds which remove ice seaward from the region more rapidly than it can be formed or advected into the region (Figure 5) [*Bromwich and Kurtz*, 1984; *Kurtz and Bromwich*, 1983, 1985]. As for the North Water, the winds are channeled by glacier valleys and can attain high velocities. The Drygalski Ice Tongue extends seaward and blocks the westward alongshore currents which would otherwise advect ice into Terra Nova Bay. Locally wind-generated currents and surface waves combine to prevent the ice, estimated by *Bromwich and Kurtz* [1984] to form at 20 cm/day, from consolidating within the polynya. *Cavalieri and Martin* [1985] have observed similar features

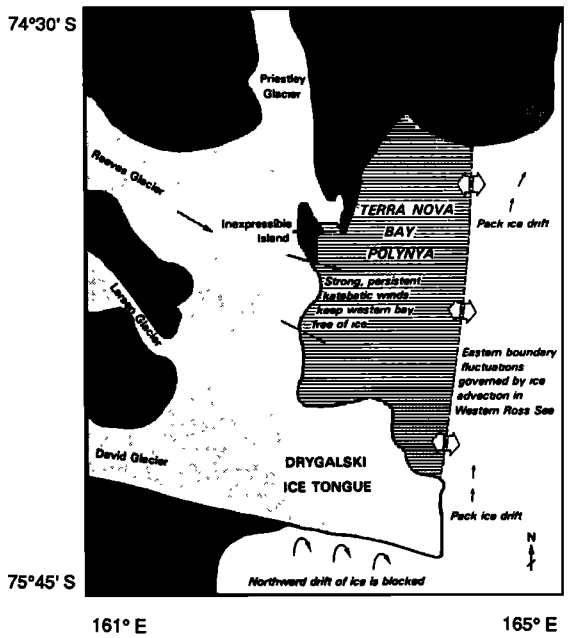

Fig. 5. Schematic chart of Terra Nova Bay polynya, showing the direction of katabatic wind influence and the offshore sweep of the sea ice [*Bromwich and Kurtz*, 1984].

along the Antarctic Wilkes Land coast using satellite imagery. *Zwally et al.* [1985] have computed brine rejection based on ice formation within such Antarctic coastal polynyas. Their estimates demonstrate that salinity of Antarctic continental shelf water is significantly increased through this mechanism.

Siberian Shelf Polynyas

Time series of open areas of polynyas in the vicinities of Nova Zemlya, Severnaya Zemlya, and Franz Josef Land were derived from passive microwave satellite data by *Martin and Cavalieri* [1989] through the winters of 1978–1982. Coastal weather station data were then used to estimate surface heat loss, ice production, and brine rejection. From oceanographic data it was inferred that shelf polynyas are capable of producing 20–60% of the Arctic Intermediate Water.

3. SENSIBLE HEAT POLYNYAS

In a sensible heat polynya, sufficient oceanic heat is made available to the water surface to prevent ice from forming. This upward heat transfer can occur through vertical mixing of heat from deeper water or through upward advection of heat by such mechanisms as wind-driven upwelling, making sufficient oceanic heat available to erode the underside of the ice cover. Once the polynya opens, the surface rapidly loses heat to the atmosphere. The source of warmer water may be sufficient to prevent freezing. If it is not, the currents which are responsible for advection may also carry away newly formed ice crystals before they can form a continuous ice cover. In this case, both sensible and latent heat transport combine to maintain the existence of the polynya. A secondary mechanism for retarding ice formation in a sensible heat polynya might be a shift in radiation balance caused by the presence of clouds enhanced by the moisture source. Unlike latent heat polynyas, sensible heat polynyas are not sites of large net winter ice production and consequent brine addition to the underlying water column. However, significant cooling of the water column does occur.

The Whaler's Bay Polynya

A good example of an Arctic sensible heat polynya occurs in winter in the "Whaler's Bay" region along the northern coastline of Svalbard (Figure 6). The regional oceanography is dominated by vigorous mixing between warm Atlantic water flowing northward in the West Spitsbergen Current and colder ambient waters. *Aagaard et al.* [1987] have computed a heat loss of more than 200 W/m^2 from the 100- to 200-m warm water which if provided directly to the upper layer can contribute significantly to preventing ice formation. *Perkin and Lewis* [1984] computed an order of magnitude smaller upward heat flux in the same region for ice covered conditions. Presence or absence of an ice cover in this region is hypothesized by *Aagaard et al.* [1987] to contribute to observed variability in regional water temperature and salinity characteristics.

Polynyas in the Canadian Archipelago

Atlantic tides enter the Arctic Basin through the gap between Greenland and Norway and propagate under the ice to the Beaufort Sea, where the relatively small tides are out of phase with the large tidal surges of northern Baffin Bay. Tidal currents are especially strong in the complex intervening channels among the islands of the Canadian archipelago. In the western archipelago there is comparatively little brine input to the upper layers, allowing the warm underlying Atlantic Water to advect along the channels [*Melling et al.*, 1984]. Vigorous mixing at constrictions in these channels brings Atlantic water to the surface, melts the ice, and gives rise to several groups of small polynyas, notably in Hell Gate, Cardigan Strait, Penny Strait, and Queen's Channel [*Markham*, 1981] (Figure 3). One of these was intensively studied with respect to oceanographic and meteorological processes [*Topham et al.*, 1983; *Smith et al.*, 1983; *den Hartog et al.*, 1983]. This study will be discussed in a later section.

The Okhotsk Polynya

A prominent polynya has been observed in the central sea of Okhotsk using satellite imagery [*Alfultis and Martin*, 1987]. This polynya appears to be largely wind driven. Oceanographic data have been interpreted by Alfultis and Martin to mean that upwelling associated with a submarine bank provides oceanic heat to maintain the polynya. A separate polynya which occurs along the northern coastline of the Sea of Okhotsk is mechanical in origin and similar to those occurring in the northern Bering Sea. Alfultis and Martin estimate that significant quantities of dense shelf water are produced through cooling and brine rejection in this coastal polynya.

Weddell Sea Polynya

Probably the best known, if not the best documented, sensible heat polynya is the Weddell Sea polynya (Figure 7). This feature appeared in the winter Weddell sea ice cover during several years in the late 1970s [*Carsey*, 1980; *Crawford and Parkinson*, 1981; *Zwally et al.*, 1985]. Detected by satellite imagery, the polynya appears to have been due to large-scale divergence and concomitant upward transport of

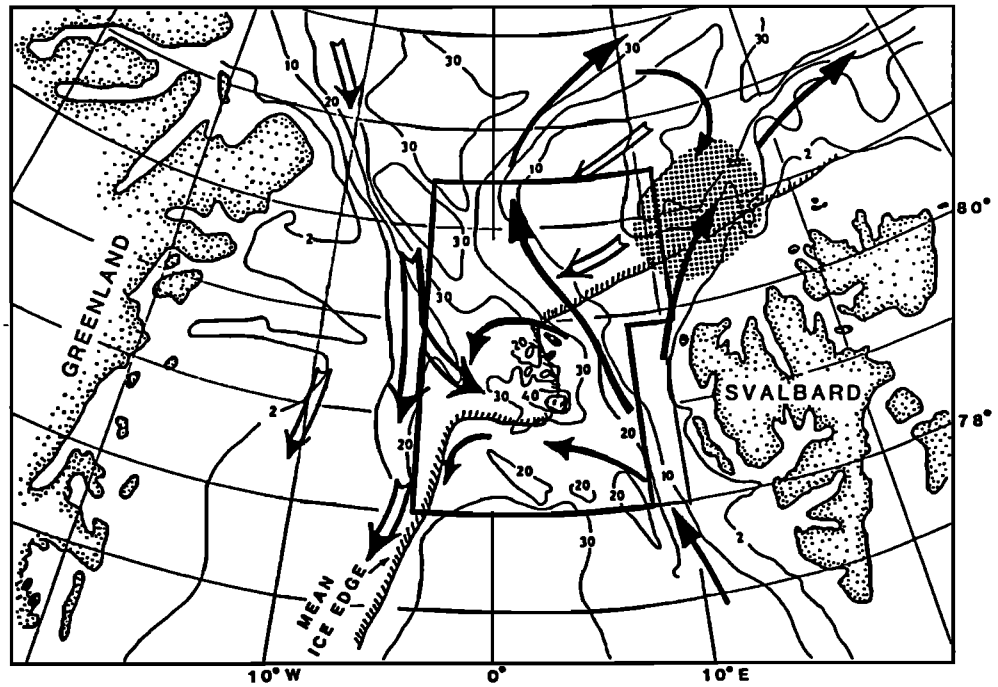

Fig. 6. Location of Whaler's Bay polynya (shaded area) showing path of warm Atlantic water (dark arrows) which provides oceanic heat to prevent ice formation, and of ice drift (open arrows). Contours are bathymetry in hundreds of meters (adapted from *Johannessen et al.* [1987]).

oceanic heat sufficient to prevent ice formation. This has been verified by numerical models [*Martinson et al.*, 1981; *Parkinson*, 1983; *Motoi et al.*, 1987] and is consistent with what is known of the regional oceanography [*Gordon and Huber*, 1984]. The Weddell polynya is presumed possible because of the very low vertical water column stability which typifies much of the southern ocean and allows vertical mixing in the central basin to transport heat from deeper layers to the surface. Conversely, deeper water in the Weddell Sea is cooled by such a mechanism, contributing to large-scale water mass modification and formation of Antarctic Bottom Water. The situation in the Weddell Sea provides a marked contrast to that of the Arctic Ocean, where vertical mixing typically does not extend downward to the warm water layer because of relatively high stability caused by a basinwide halocline/pycnocline.

Two considerably smaller deep-ocean polynyas, similar to the Weddell polynya, were observed east of the Weddell polynya region in winter 1980 using satellite imagery. These features probably resulted from the same mechanisms which caused the Weddell polynya. *Comiso and Gordon* [1987] attributed these features to oceanic heat made available to the surface waters by localized upward doming of the pycnocline as indicated by historical oceanographic data from the region. They also hypothesize that these polynya events are terminated when favorable atmospheric conditions relax, which attenuates the vertical convection carrying heat to the surface, and the polynyas are invaded by sea ice from the sides.

As compared with the latent heat polynyas, whose primary oceanographic effect is introduction of brine into the water column, sensible heat polynyas are sites of very large sea-air heat fluxes which remove large quantities of heat from the water. Some typical heat flux values are presented in Table 1.

4. SURFACE HEAT BUDGET

The atmospheric processes associated with polynyas and leads are basically the same. Extreme sea-air temperature differences of 20°–40°C are common in winter. Upward sensible and latent heat fluxes depend mainly on air temperature, wind speed, and fetch. The sensible heat flux in air is several times larger than the latent heat flux because of the relatively low specific humidity of saturated air in contact with sea water at the freezing point. Radiative transfers may be influenced by fog, which is characteristic of very cold air over a water surface.

The surface heat balance is expressed by

$$H + LE - Q_s(1 - A) + Q_L = H_w + L_f F \qquad (1)$$

where H and H_w are the upward sensible heat fluxes in air and in water, L and L_f are the latent heats of evaporation and of fusion, E and F are the rates of evaporation and freezing at the surface (mass per unit area per unit time), Q_s is the incident solar radiation, A is the surface albedo, and Q_L is the net upward longwave radiation. Positive values indicate upward heat transport. Terms on the left-hand side represent atmospheric processes, while those on the right-hand side occur in the water. All these terms are vastly different for the open water of a lead or polynya than for the surrounding ice and snow surface.

Turbulent Fluxes in Air

The surface water temperature T_s remains close to the freezing point. The sensible heat flux in air depends mainly on the ambient air temperature and wind speed and also on fetch from the upwind edge [*Smith et al.*, 1983; *Andreas and Murphy*, 1986; *Smith*, 1988]. In winter, fluxes of several hundred watts per square meter from open water surfaces

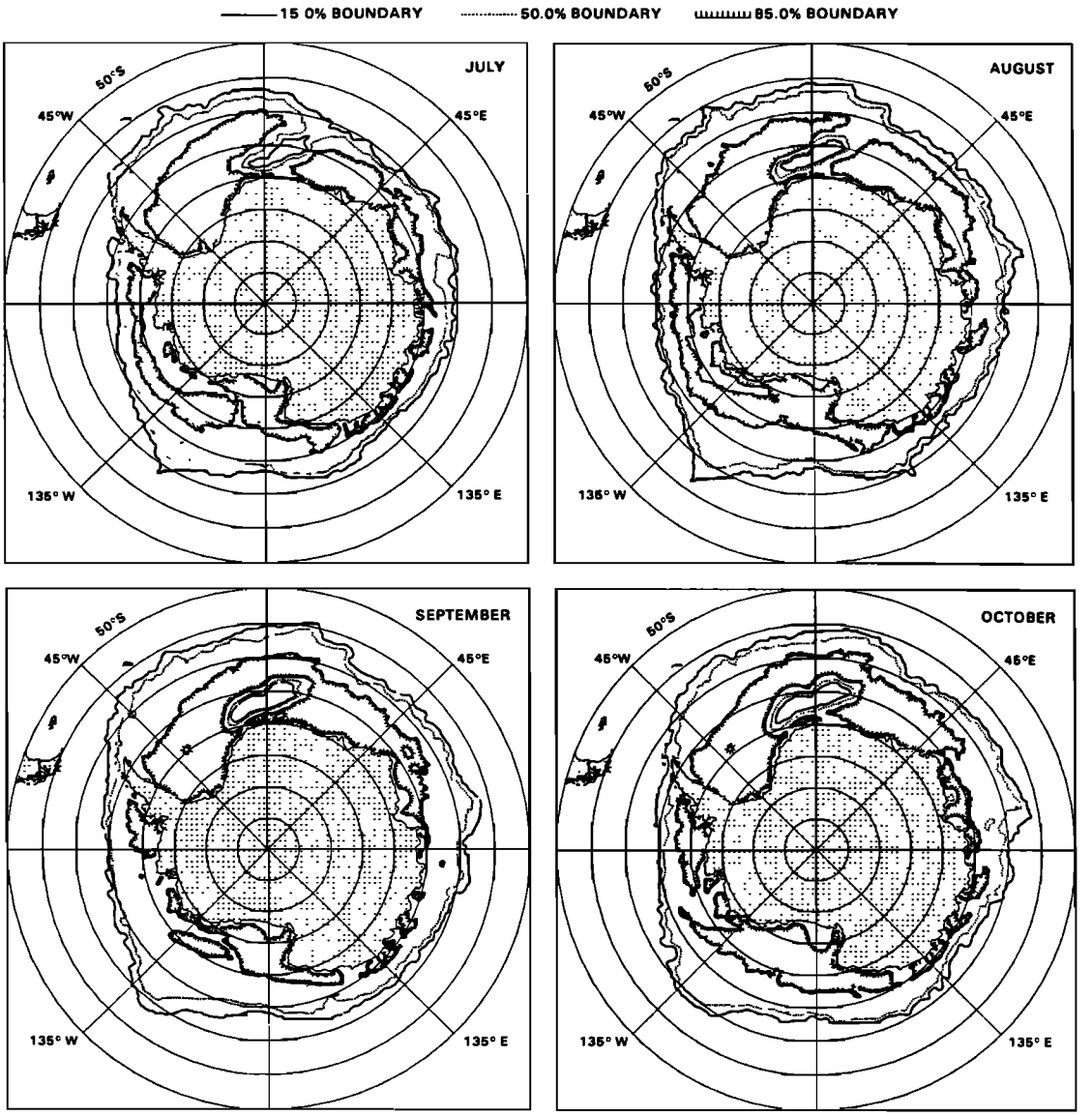

Fig. 7. Monthly mean Antarctic ice concentration contours for September and October 1984, based on ESMR brightness temperatures, showing the Weddell Sea polynya just offshore and centered on the Greenwich meridian [*Zwally et al.*, 1985].

are typical (Table 1) while in summer, with the air temperature near the freezing point of seawater, the sensible heat flux is much smaller. The evaporative heat flux *LE* is determined mainly by the wind speed, since specific humidity at the surface is fixed at the saturation value at the freezing point, while the specific humidity of cold overlying air is much smaller regardless of the relative humidity. Thus the specific humidity difference is nearly constant. The latent heat flux is typically about one-quarter as large as the sensible heat flux in winter [*den Hartog et al.*, 1983], and in summer it becomes small if the ambient humidity is high. At air temperatures below −20°C an ice fog forms as warm, moist surface air mixes upward and cools. Plumes of warm, moist air have been observed by airborne lidar to rise from large leads to heights of up to 4 km [*Schnell et al.*, 1989]; this surprising finding will open a new and fascinating area of research.

Radiative Fluxes

Shortwave radiation is zero or negligible through the Arctic winter but amounts to several hundred watts per square meter in spring and summer. Ice and snow have a high albedo, so that most incoming solar radiation is reflected, while open water has a low albedo [*Payne*, 1972] and most radiation is absorbed in the upper layers of the water. Leads and polynyas in the summer act as "windows," and absorbed solar heat promotes further melting around their edges [*Hall and Rothrock*, 1987]. Accelerated melting as more and more open water absorbs solar heat can contribute to the breakup of sea ice.

The net longwave radiation is the difference between the nearly constant upward radiation (determined by surface temperature and the emissivity of water) and the more variable downward component, which depends on the temperature and emissivity of overlying water vapor, fog, or cloud layers. The net longwave radiation loss is typically not much greater than 100 W/m^2. Snow cover over ice in winter is typically far below the freezing point and has a smaller longwave heat loss. In summer the snow or ice is near the freezing point, and there is little longwave contrast between ice and water surfaces.

TABLE 1. A Compilation of Air-Sea Heat Fluxes and Ice Formation Rates for Selected Polynyas

Source	Sea-Air Heat Flux, W/m^2	Time Periods	Mean Ice Production, m/day	Region	Type
Kurtz and Bromwich [1985]	713	May–Sept.	0.31	Terra Nova Bay*	latent heat
Cavalieri and Martin [1985]	160–450	June–Sept.	0.05–0.13	Wilkes Land coast†	latent heat
Schumacher et al. [1983]	535	65-hour event	0.17	St. Lawrence Island	latent heat
Pease [1987]	150–500	Feb. 1982, 1983, 1985	···	St. Lawrence Island	latent heat
den Hartog et al. [1983]	329	March 1980	···	Penny Strait	both
Alfultis and Martin [1987]	446	Dec.–March 1978–1982	0.13	Sea of Okhotsk	sensible heat
Steffen [1985*b*]	180	Nov.–March	···	North Water‡	both

*High value is due to large sensible heat flux/ice formation rate.
†Computed at three sites: maximum 4-day mean exceeded 800 W/m^2 at one site.
‡Includes areas covered by thin or new ice.

Processes in Water

The heat transport in the water depends on the regional oceanography and can in some areas be as large as any of the other terms. If all of the upper water layer is at the freezing point, heat transport by the water is negligible, and any loss of heat results in freezing.

The final term in (1), the freezing or melting rate, is a residual of the other terms and acts to hold the water close to the freezing point at least until such time as the surface freezes over or the surrounding ice pack breaks up. New frazil ice formed at the surface is in the form of platelets or spiny ice crystals.

The freezing process is accompanied by rejection of brine and acts as a source of salt. The salinity of new ice depends on the freezing process, and brine drainage can continue long after the initial freezing. Near the freezing point the density of seawater depends almost entirely on salinity, and dense water formed at the surface sinks [*Kozo*, 1983]. Depending on the strength of the pycnoline at the base of the upper mixed layer, this convection can be confined to the upper layer or can penetrate to become a source of deep or bottom water masses.

Heat Budget Studies

Even before intensive studies were carried out, surface heat budgets demonstrated the major potential of leads and polynyas in creating water masses and in ice formation. *Badgley* [1966] estimated a monthly heat budget (equation (1)) for both ice covered and open areas of the Arctic Ocean using meteorological data obtained by a University of Washington group during and after the International Geophysical Year. The largest net surface heat loss from open water (left-hand side of (1)) was 884 W/m^2 in February, when heavy pack ice lost only 1.7% as much heat. The largest net heat gain was 237 W/m^2 for open water in June, when ice-covered areas gained 9.7% as much heat.

Vowinckel and Orvig [1973] used an energy budget model [*Vowinckel and Orvig*, 1971] based on their earlier work [*Vowinckel and Orvig*, 1966; *Vowinckel and Taylor*, 1965] to calculate daily average budgets for pack ice and for open water in the central Beaufort Sea using 11 months of meteorological and upper air data (May 1957 to March 1958) from the Soviet drifting station North Pole 7. By far the largest term was the sensible heat flux from open leads and polynyas. The ice surface in winter was cooled below ambient air temperature by longwave radiative losses and the sensible and latent heat fluxes were downward. A net heat loss (left-hand side of (1)) of 1073 W/m^2 in February over open water agreed remarkably well with Badgley's value of 884 W/m^2, considering that the data used were from different years and different stations. Only 0.1% of open water was needed to have a net winter heat loss equal to that from all the remaining area. The highest monthly gain, over open water in June, was 141 W/m^2 or 59% of Badgley's value. Variation of ice thickness between 1 and 4 m was estimated to be of minor consequence in relation to the contrast with open water. These early heat budget studies produced plausible results, but they were based on estimates of processes which at the time had not been directly measured. They provided a stimulus and a framework for field studies as discussed in the following sections.

On the basis of low-level remote sensing flights and NOAA very high resolution and radiometer (VHRR) satellite imagery, *Steffen* [1985*a*] and *Steffen and Ohmura* [1985] have mapped the effective surface temperature of the mixture of ice and open water in the North Water region (Figure 3). Combining this with meteorological observations and the ice distribution [*Steffen*, 1986], a monthly and annual surface heat budget was calculated, ranging from a net loss of 215 W/m^2 in January to a net gain of 226 W/m^2 in June, with a yearly mean loss of 33 W/m^2. In winter the largest component of heat loss is sensible heat flux (139 W/m^2 in January), while in summer the largest gain is absorbed solar radiation (254 W/m^2 in June).

5. ICE CONDITIONS IN LEADS AND LATENT HEAT POLYNYAS

The mean ice production in a latent heat polynya can be dramatic (Table 1) when contrasted with typical rates of less than a centimeter per day for congelation beneath an existing cover. The ice formed in a latent heat polynya is typically frazil because of the high levels of ambient turbulence which are generated by the winds required to maintain the polynya [*Martin*, 1981; *Weeks and Ackley*, 1982; *Bauer and Martin*, 1983; *Omstedt and Svensson*, 1984; *Pease*, 1987]. Crystal size and orientation depend upon the turbulence level; small randomly oriented crystals form for high wind speed regimes, and large flat or spiny crystals form for low wind speed regimes [*Weeks and Ackley*, 1982]. Initially, these crystals are loosely joined, and a light wind or current can sweep the crystals to the downstream ice edge, leaving the majority of the surface of a lead or polynya ice free. A strong

current can sweep the frazil ice under the pack ice edge, where the frazil ice eventually attaches itself to the underside of the pack ice at places and times where the currents are weaker. In calm conditions, supercooling of the water surface has been observed [*Katsaros*, 1973]. At higher wind speeds, less supercooling of the water column is required to initiate ice production at the beginning of a freezing event, and frazil forms at greater depths throughout the event [*Omstedt and Svensson*, 1984].

For wind speeds above some threshold (probably 7–8 m/s) ice production takes on a physically structured aspect. Langmuir circulations start organizing the frazil ice into rows at the surface [*Pease*, 1987]. The frazil in these rows is swept downwind at speeds higher than those of the receding pack ice because of Stokes drift from wind waves and/or because the downwind surface current is positively enhanced at the Langmuir surface convergence lines. Frazil collection at the downwind edge of the polynya for lower wind speeds is uniform to some depth along the receding pack ice. The depth of this collection depends largely upon the thickness of the pack ice, the wind speed and the air temperature [*Bauer and Martin*, 1983]. At high wind speeds and very low air temperatures, however, the frazil mats start to form floes before the frazil can be fully compacted against the receding ice floes, leaving the young ice perforated with holes 1–5 m in diameter. Although there is still a large percentage of open water within this perforated field of floes, energy transmission to the atmosphere and salt enhancement of the ocean are greatly attenuated compared with values in the open portion of the polynya because ice production in low-turbulence regimes is much less efficient than where turbulence is fully developed and Langmuir cells are present. The type of ice crystals formed changes to larger, flat plates in this low-turbulence region, with the plates oriented horizontally in the surface holes and forming columnar ice under the solidified frazil mat [*Weeks and Ackley*, 1982].

The total brine trapped in the ice is a function principally of air temperature. More brine is trapped during rapid freezing than during slow freezing. Consequently, salt content of new ice varies from about 6 ppt for higher air temperatures to 10–15 ppt for lower temperatures. Drainage of brine continues, so that multiyear ice typically has only 0–1 ppt salt [*Weeks and Ackley*, 1982; *Untersteiner*, 1986].

6. Review of Selected Experiments

Artificial Lead Experiments

In March 1960, *Miyake* [1961] measured wind and temperature profiles upwind and downwind of an artificial lead formed by pumping seawater onto shorefast ice near Barrow, Alaska. The instrumentation deployed for this pioneering experiment served as a model for several later experiments. Wind and temperature profile masts were placed at the upwind and downwind edges with cables running to data loggers in a shelter. The experiment was designed to test theoretical predictions of growth of the height δ of the internal boundary layer. For a wind speed of 9 m/s and a temperature difference of 17.5°C, the predicted internal boundary layer heights were 26 and 68 cm at 10 and 20 m fetch, respectively, while measured heights were 35 and 140 cm, about twice as large. The theory was qualitatively successful in predicting an accelerating growth of the internal boundary layer with fetch in highly convective conditions.

Miyake did not estimate surface heat fluxes, although in principle these could have been derived from his data. The integral method determines the amount of heat required to raise the temperature of the air column as it crosses a lead,

$$\bar{H} = H_i + (\rho c_p/X) \int_0^{\delta} U(T - T_i)\, dz \qquad (2)$$

where H_i is the heat flux over the upwind ice, ρ and c_p are the density and specific heat of air, X is the fetch over open water, U is the wind speed, and T_i and T are the upwind and downwind temperatures at height z. *Badgley* [1966] reported a sensible heat flux of 294 W/m^2 and a latent heat flux of 48 W/m^2 in a similar artificial lead experiment near Barrow in March 1962 with an air temperature of −27°C and a wind speed of 3.2 m/s.

The AIDJEX Lead Experiment

The AIDJEX (Arctic Ice Dynamics Joint Experiment) Lead Experiment (ALEX), February 23 through April 10, 1974, investigated small-scale meteorological and oceanographic processes associated with leads in pack ice near Barrow, Alaska. The experiment plan [*Paulson and Smith*, 1974] called for rapid deployment of five instrumented huts, measuring equipment and personnel by helicopters and fixed-wing aircraft. Seven helicopter flights and six fixed-wing aircraft flights were required for deployment. The processes of primary interest were sensible, latent, and radiant heat loss to the atmosphere, and sinking of convective plumes of saline water formed by freezing and brine rejection at the surface. Sensible and latent heat fluxes were studied at the upwind and downwind edges by teams from Oregon State University and the University of Washington, and radiation was measured by a group from the University of Alaska. A team from the University of Washington carried out the oceanographic investigations. Since a newly formed lead could freeze over in less than a day, data gathering had to begin as soon as possible after the discovery of a lead. In spite of the simple concept, logistical problems hampered the execution of the experiment [*Heiberg*, 1974; *AIDJEX*, 1974]. Helicopter range limited deployment to within 30 miles of Barrow, and often no suitable lead was found within the search radius.

Radiation. The radiation program was moderately successful. *Holmgren and Weller* [1974] reported net radiative fluxes of +26 to −101 W/m^2, but they found that [*Holmgren and Weller*, 1974, p. 151] "at the open leads . . . virtually nothing could prevent frost from forming on the instruments within a few minutes." This prevented the gathering of the desired data set.

Atmospheric boundary layer. *Andreas et al.* [1979] and *Andreas* [1980] measured sensible heat fluxes by three methods. The integral method was usable in 57 of 76 reported runs; heat flux was deducted from differences between upstream and downstream profiles. The profile method, which deduces heat flux from vertical temperature and wind gradients, gave generally similar heat flux values although preliminary analysis [*Lindsay*, 1976] had suggested higher values.

The eddy flux method directly determines the vertical heat

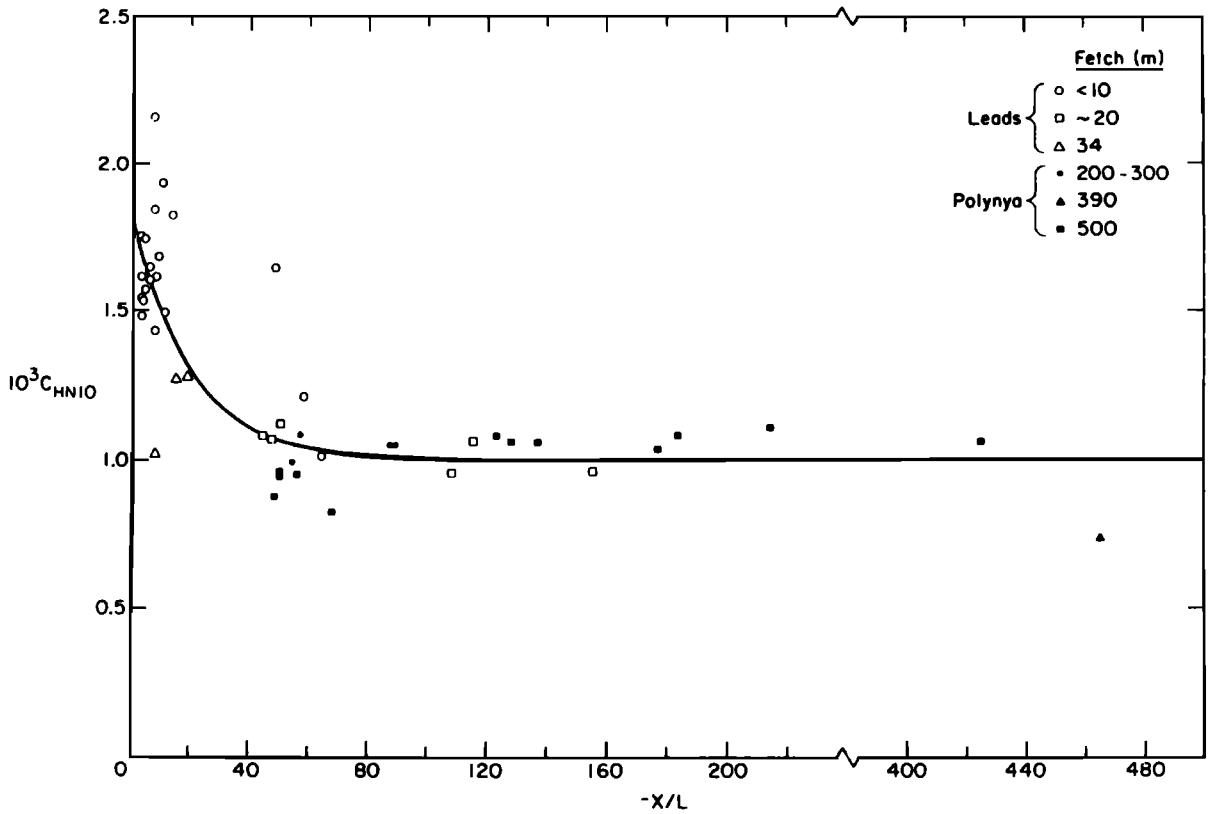

Fig. 8. The sensible heat transfer coefficient for neutral stability as a function of dimensionless fetch [*Andreas and Murphy*, 1986].

flux H past a fixed measuring point by correlating fluctuations in turbulent temperature t and vertical wind u_3

$$H = \rho c_p \overline{tu_3} \tag{3}$$

The overbar denotes a time average, typically over periods of 20–60 min. Rime ice formation made the response of thermocouple thermometers too slow for eddy flux determinations. The turbulence spectra and other properties of turbulence were reported by *Andreas and Paulson* [1979]. *Andreas* [1980] derived two empirical formulas which describe the variation of heat flux with fetch, and *Andreas et al.* [1981] ingeniously used spikes in signals from hot-film anemometers to derive similar formulae for the flux of condensate droplets. The empirical fetch averaged heat flux is [*McBean*, 1986]

$$\bar{H} = \frac{1}{X}\int_0^X H\,dx$$

$$= 2.5 \times 10^{-3}\rho c_p U_2(T_s - T_2) + 1100\kappa(T_s - T_2)/X \tag{4}$$

where κ is the thermal conductivity of air. The influence of the last (fetch dependent) term in (4) is to add a flux of 0.27 $(T_s - T_2)$ W/m of lead length to the total sensible heat loss, regardless of lead width. This term contributes only 8% of the total flux for a fetch $X = 33$ m.

For polynyas and leads wider than about 30 m, where fetch dependence is not critical, the "bulk formula" for sensible heat flux is

$$H = \rho c_p C_T \quad U_r(T_s - T_r) \tag{5}$$

where C_T is a "bulk coefficient," U_r is the mean wind speed and T_r is the air temperature at the reference height r. *Andreas and Murphy* [1986] and *Andreas* [1988] combined the ALEX eddy flux data with polynya data from *Smith et al.* [1983] to recommend a bulk coefficient for neutral conditions (subscript N) which depends on fetch X (Figure 8)

$$10^3 C_{TN10} = 1.0 + 0.8 \exp 0.05X/L \tag{6}$$

where the Obukhov length is

$$L = \rho c_p u^{*3}(T + 273°)/gKH \tag{7}$$

in which $g = 9.8$ m/s^2 is the acceleration of gravity, $K = 0.4$ is the von Kármán constant, and $u^* = (\tau/\rho)^{1/2}$ is the friction velocity (τ is wind stress). Flux gradient formulas can be applied to iteratively solve for C_T [e.g., *Smith*, 1988], which is substantially larger than C_{TN} in the highly unstable conditions over a polynya.

Ocean boundary layer. During the third deployment of the AIDJEX Lead Experiment a current jet of 5–8 cm/s was observed deep in the mixed layer (Figure 9), and a shore lead some tens of kilometers away was suggested as a possible source. During the fourth deployment a jet of between 3 and 12 cm/s was found with a velocity maximum 0.5 m beneath the ice (Figure 9), directed toward an area of open ponds in a refreezing lead 200 m away. This jet was inferred to be supplying fluid to replace material lost to sinking of brine in convection cells. During the second deployment the lead closed and formed a pressure ridge before data could be obtained. None of the four oceanographic deployments produced any data in the immediate vicinity of an open lead.

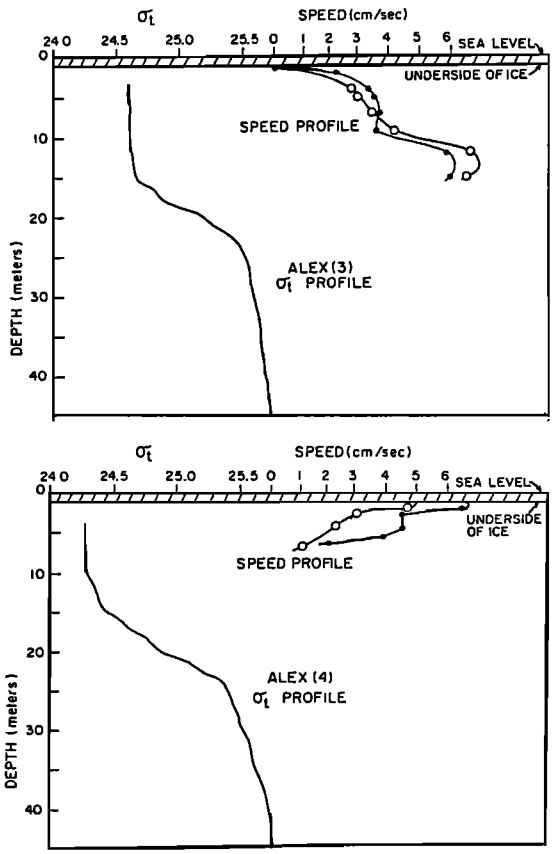

Fig. 9. Current jets and velocity profiles observed during ALEX [*Smith*, 1974]. See text for discussion.

The ultimate goal to "provide information on the mechanical and thermodynamic processes active in the mixed layer attributable to the presence of open leads" was not accomplished [*Smith*, 1974, pp. 125 and 132].

The Dundas Island Polynya Experiment

Topham et al. [1983] studied heat exchange processes at the Dundas Island polynya, a small (1 km) "sensible heat" polynya in the Penny Strait area (Figure 3). The study was conceived to provide information on the heat exchange from moderate scale open water features as an aid to estimation of heat loss from the Arctic Ocean, and also to identify the oceanographic phenomena responsible for the regular occurrence of a polynya at a particular site. The experiment, performed in March and April 1980, was a cooperative venture with teams from three Canadian agencies: groups from Bedford Institute of Oceanography and Atmospheric Environment Service concentrated their efforts on the surface heat exchange and atmospheric boundary layer modification, while a group from the Institute of Ocean Sciences provided a picture of oceanographic conditions in the area.

Oceanography. The polynya was located 1 km north of Dundas Island in Pioneer Channel, which is about 10 km wide and is bridged at its narrowest point by a sill 40 m deep. The water depth in the polynya itself was 30 m, except near its eastern edge where it shallowed to 18 m. Five current meters were deployed across the channel. At the polynya the currents followed a semidiurnal tidal cycle with flow mainly to the east, peaking at 1.2 m/s during spring tides and half of this value during neap tides. Surface water temperatures were as much as 0.2°C above freezing despite extremely low air temperatures. These elevated surface water temperatures were due to advection and upward mixing of warmer water originating from the warm Atlantic water layer in the Arctic Ocean [*Melling et al.*, 1984]. The high upper layer stratification within the Canadian archipelago retards upward flux of heat out of the Atlantic water until it enters the channels of the archipelago, so that the heat is adequate to prevent ice formation when mixed upward by strong tidal currents.

Like the St. Lawrence Island polynya in the Bering Sea, this feature was not steady state but varied in size, expanding during periods of spring tidal currents which mixed heat upward more vigorously and also when air temperatures were higher. Analysis of satellite photographs and data from a nearby weather station (Resolute Bay, Northwest Territories) showed how the size of the polynya responds to air temperatures: in 1977, 1979, and 1982, episodes of −40°C temperatures in February were followed by freezing over of the polynya during neap tides in early March. The 1979 freeze-up terminated a field experiment shortly after the deployment phase. In 1978, 1980, and 1981, February temperatures, at times rising above −20°C, were followed by persistence of open water throughout the spring.

Surface fluxes. Sensible sea-air heat flux and surface wind stress were determined by eddy correlation, profile, and integral methods. The eddy correlation method is a direct measure of the wind stress,

$$\tau = -\rho\overline{u_1 u_3} \tag{8}$$

and the turbulent sensible heat flux H (equation (3)) at a fixed point in the atmospheric surface layer, where u_1 and u_3 are the downwind and vertical components of the turbulent wind velocity fluctuations. In this experiment a sonic anemometer was operated at a height of 4.4 m at the downwind edge of the polynya [*Smith et al.*, 1983]. Twenty-three data runs were analyzed to obtain sensible heat flux and wind stress for wind speeds of 2–9 m/s and sea-air temperature differences of 20°–32°C. Heat flux coefficients C_T measured over the polynya confirmed the validity of stability-dependent coefficients tabulated by *Smith* [1981, 1988] for computation of the heat flux by the bulk formula (5), as did heat fluxes from *Andreas et al.* [1979] at a fetch of 20 m over a lead. In very low wind conditions an alternative empirical formula based on free-convection scaling was used whenever it gave a larger heat flux than (5). A comparison of estimated and measured sensible heat fluxes gave agreement within 10% in the majority of cases. Wind stress on the polynya was comparable to that on other water surfaces in highly unstable stratification; the structure of atmospheric turbulence for unstable stratification was also investigated.

Determination of the sensible heat flux by the integral method (equation (2)) did not give reliable heat flux values because there was a wide range of heights with small, poorly resolved temperature differences. Fluxes estimated from wind and temperature profile gradients were also considered less reliable than the turbulent flux. The bulk formula for the evaporation rate is

$$E = \rho C_E U_r (q_s - q_r) \tag{9}$$

where q_s and q_r are the saturation specific humidity at the surface temperature and the ambient specific humidity at a

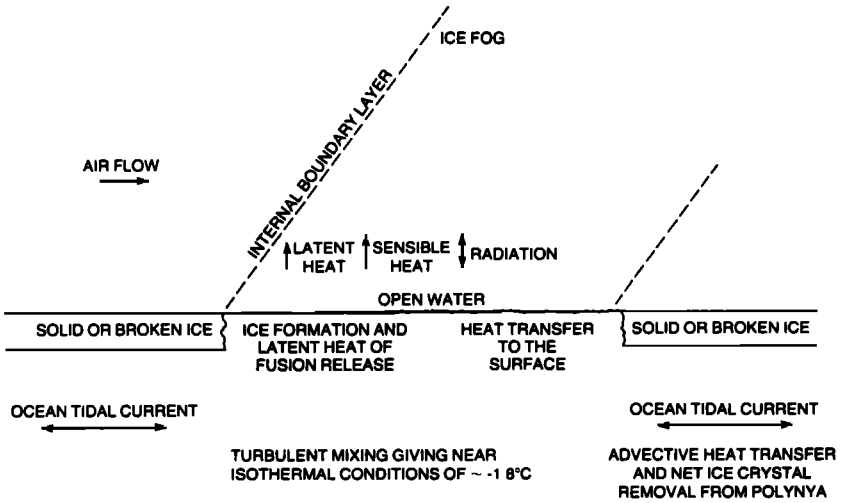

Fig. 10. Polynya heat transfer process [*den Hartog et al.*, 1983].

reference height r, respectively. The latent heat flux LE is typically much less than the sensible heat flux in Arctic conditions but has not to our knowledge been directly measured over leads and polynyas. *Smith* [1988] recommends $C_E = 1.2\ C_T$ on the basis of measurements over oceans in temperate regions; *Andreas and Murphy* [1986] recommend $C_E = C_T$.

The total heat loss from the water was measured directly by a technique adopted from *Lebedev* [1968]. A shallow fiberglass tray was floated flush with the surface of a thin shelf of newly formed ice near the upwind edge of the polynya. Since the water was initially at the freezing point, virtually all heat loss from the surface of the tray was accompanied by formation of ice crystals. From the salinity of the liquid water before and after exposure, heat losses of 384–437 W/m^2 were reported on four occasions, on days when the average heat loss was estimated at 220–350 W/m^2. The trays were at the upwind edge of the polynya, where larger sensible heat losses are expected [*Andreas and Murphy*, 1986] and where the absence of ice fog permits larger longwave radiative losses. Because of the strong currents and often active ice dynamics at the edge, there were limited opportunities to deploy the flux trays.

Heat budget. The heat budget of the polynya (Figure 10) was calculated from March 12 to April 1, 1980 [*den Hartog et al.*, 1983]. A bulk formula (equation (5)) with coefficients from *Smith* [1981, 1988] was used to calculate sensible heat flux from wind and temperature records. Evaporation was estimated using the same coefficients. Longwave radiation was estimated by using the Stefan-Boltzmann law for the upward component and Paltridge's formula [*Paltridge and Platt*, 1976, p. 140] for the downward component. Shortwave radiation measurements were obtained from Resolute, Northwest Territories, and albedo was made to vary as a function of air temperature to allow for the influence of ice fog, which was frequently observed when temperatures were below −30°C.

Sensible heat flux was the dominant term in the surface heat budget, averaging 204 W/m^2 for the entire period and exceeding 400 W/m^2 on the coldest day (Figure 11). The latent heat flux was about 25% of the sensitive heat flux. The net heat loss for the period was estimated to be 329 W/m^2.

7. MODELS OF PROCESSES AT POLYNYAS AND LEADS

Atmospheric Boundary Layer Models

Lo [1986] extended the two-dimensional internal boundary layer model of *Taylor* [1970] to fit the case of a lead or polynya. He used a closure in which turbulent dissipation is a function of turbulent kinetic energy and of a "dissipation length" proportional to height. The transition from a cold, dry, rough upwind ice surface to the warmer, wetter, and smoother surface of the open water requires inclusion of both heat and vapor, as well as the usual momentum, mass, and turbulent energy equations. The solutions are given in a graphic form (e.g., Figure 12), and modeled temperature and velocity profiles show reasonably good agreement with field data. Taking the roughness length over the water to be $z_1 = 10^{-4}$m, this figure runs from heights of 1 mm to 100 m and fetches from 0.1 m to 1 km. At fetches of $X \leq 1$ m the surface heat flux remains constant ±7%, and the height of the constant flux layer continues to grow, reaching 10 m for a fetch of 1 km. These results should be equally applicable to atmospheric boundary layers over both leads and polynyas.

Shreffler [1975] also developed a two-dimensional numerical model of the internal boundary layer over a lead. His eddy diffusivity model incorporated a thin surface sublayer with heat and momentum transfer mainly by molecular diffusion and viscosity. This model treats heat as a passive scalar, thereby ignoring the substantial influence of buoyant convection on the boundary layer. Nevertheless, the variations of heat flux and other parameters with height and fetch in the internal boundary layer are qualitatively similar to those of *Lo* [1986].

Ocean Circulation Model

Kozo [1983], following *Schaus and Galt* [1973], has developed a model for ocean circulation beneath a lead driven by buoyancy flux, freezing and brine rejection at the surface. The model is based on empirical equations for free and forced convection in atmospheric boundary layers. For conditions comparable to those observed during ALEX, the

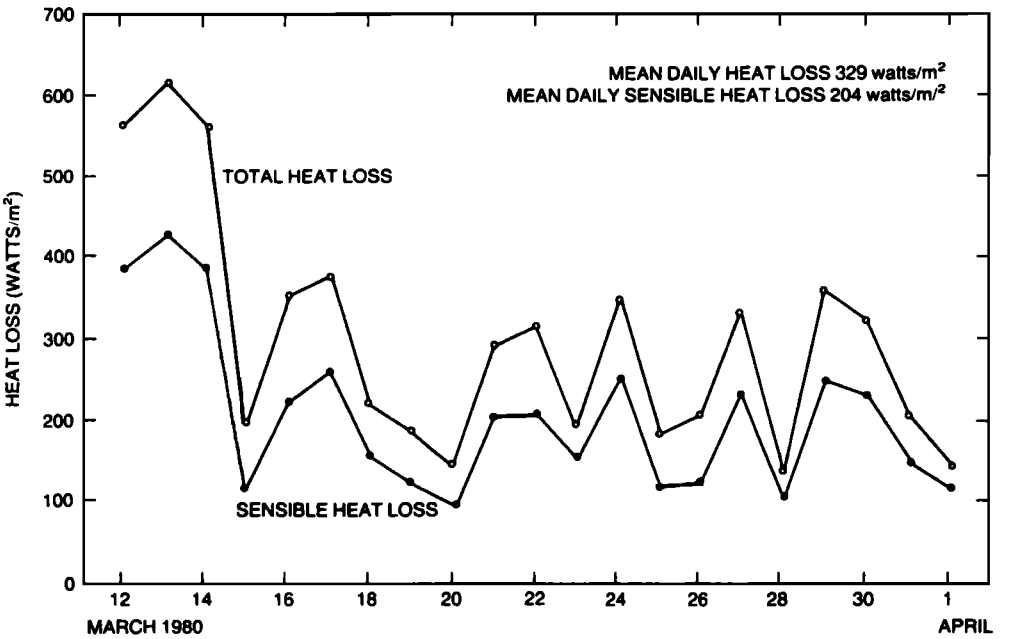

Fig. 11. Daily heat budget of Dundas Island polynya [*den Hartog et al.*, 1983].

modeled circulation in the "free convection" case (zero ambient geostrophic current) agrees in a general way with the velocity and direction of current "jets" observed by *Smith* [1974] in ALEX (Figure 13). Adding a current of 5 cm/s along the lead or 2.5 cm/s across the lead (or equivalently an ice drift rate relative to the water at the base of the upper mixed layer) reduced the convective pattern or moved it entirely to the downstream side. For a 5-cm/s cross flow there is minimal perturbation of the mean flow; in this case the salt flux is mixed downward almost entirely by turbulent shear flow. This model can serve as a basis for the planning of future experiments to study convection patterns around leads but must at this time be viewed as untested.

Wind-Driven Coastal Polynya Models

Polynyas form in the northern Bering Sea in the lee of St. Lawrence Island and the Seward Peninsula during periods of strong offshore winds. In this area the drift of winter pack ice is mainly wind driven so that when the winds are light or onshore, the polynyas freeze over. The northern Bering Sea polynyas represent the simplest possible case wherein the polynya is maintained by winds or currents in combination with a physical feature such as a coastline, channel constriction, or fixed glacial ice. It is presumed that air temperatures are sufficiently low that freezing occurs. The winds or currents then remove the ice as it is formed, while the physical features prevent existing ice from entering the region. The case for a straight coastline with offshore winds has been modeled by *Pease* [1987], who explores the relationship among environmental parameters such as wind speed and air temperature and the growth rates and maximum sizes of polynyas.

Since the upper water layer is uniformly at the freezing point, the sensible heat flux in the water column can be neglected, and in winter the solar radiation in the polar regions is also negligible. Latent heat flux is also ignored in this study, so that the surface heat budget (1) is simplified to

$$L_f F = H + Q_L \tag{10}$$

A clear-sky formula is used to model net longwave radiation. Sensible heat flux H is estimated by a bulk formula (5) with C_T set to 2×10^{-3}. Ice at the downwind edge of the polynya is assumed to drift away at 3% of the wind speed, while frazil ice formed in the polynya is assumed to drift rapidly and solidify as new ice at the downwind edge (Figure 14). The polynya grows until accretion matches the drift rate [*Lebedev*, 1968]. The steady state width of the polynya (Figure 15) and the time to reach a steady state are functions of wind

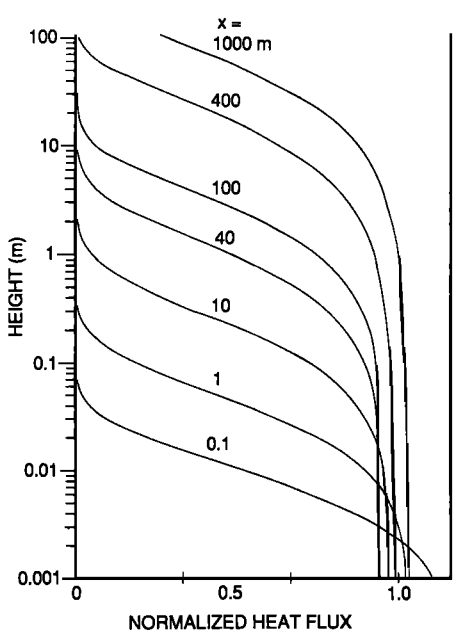

Fig. 12. Modeled sensible heat flux distribution with fetch and height, normalized to flux at 10-m height and long fetch, assuming water surface roughness length $z_1 = 10^{-4}$ m (adapted from *Lo* [1986]).

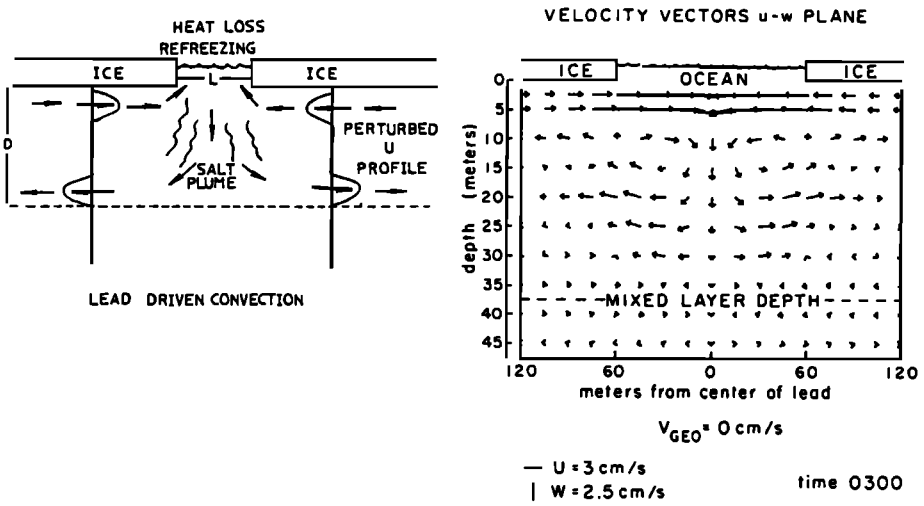

Fig. 13. Modeled convective circulation under a lead [*Kozo*, 1983].

speed, air temperature, and the frazil ice collection thickness. Since the change in polynya width with time was proportional to the difference between the advection rate of the pack ice away from shore and the production rate over the width of the polynya, the polynya width is

$$X_p = (V_i H_i / F)[1 - \exp(-tF/H_i)] \tag{11}$$

where V_i is ice velocity, H_i is collection thickness and X_p is polynya width. This function has an upper bound of

$$X_p = (V_i H_i / F) \tag{12}$$

For a "standard case" of −20°C air temperature, 20-m/s wind and 0.1-m accretion thickness, a polynya 18 km wide develops after 24 hours.

For typical winter conditions, T < −5°C and U > 5 m/s, the sensible heat flux is the dominant term in (10). The polynya width is then largely insensitive to wind speed, since ice drift rate and sensible heat loss are both proportional to wind speed. The width is almost inversely proportional to surface-to-air temperature difference, so the total heat loss is nearly independent of air temperature but proportional to wind speed. For the experimental conditions prevailing in experimental studies in 1982 (Bering-82 Experiment), 1983 (Marginal Ice Zone Experiment (MIZEX)–West), and 1985 (Arctic Polynya Experiment, or APEX), the observed widths were 10–20 km, in good general agreement with the model (15–18 km). The time to approach steady state conditions was observed to be 24–36 hours, again in good agreement with the model estimates of 23–34 hours.

Ou [1988] extended the Pease steady state model to

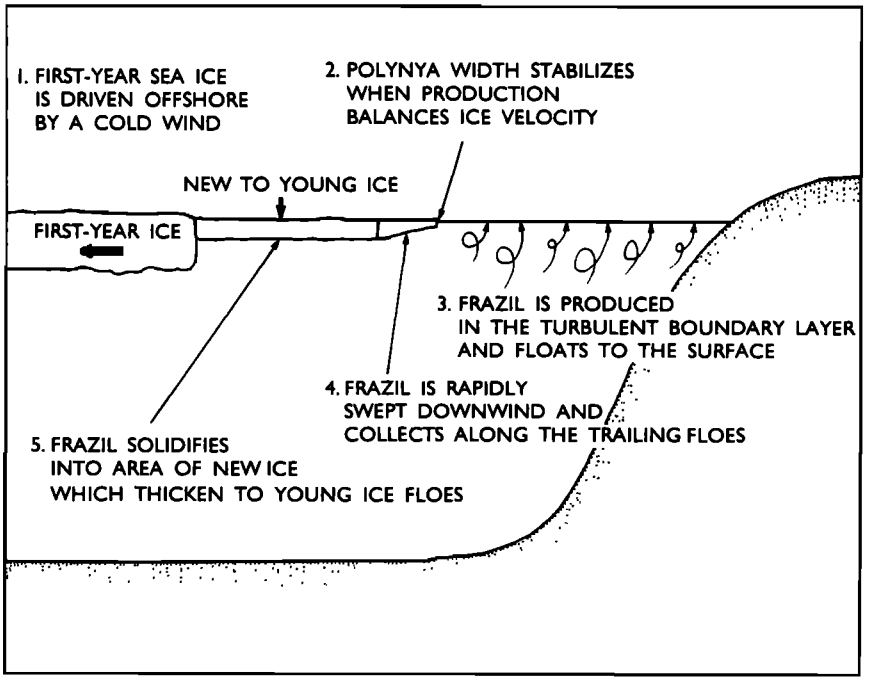

Fig. 14. Wind-driven polynya processes [*Pease*, 1987].

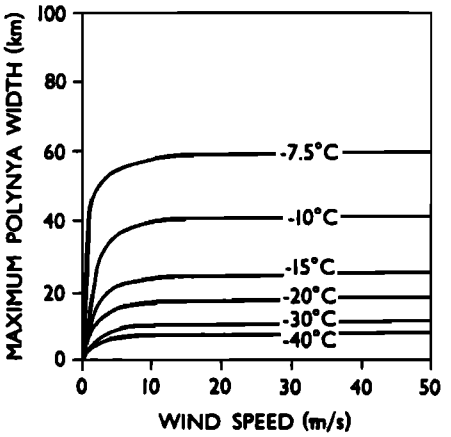

Fig. 15. Dependence of steady-state polynya width on temperature and wind speed. Note that for wind speeds above 5 m/s the size of the polynya is controlled mostly by the air temperature [*Pease*, 1987].

include time-dependent winds and temperatures and a finite drift rate for frazil ice. He found that the position of the ice edge does not respond appreciably to variations with periods shorter than the transit time for frazil ice in the polynya (typically of the order of 1 day) but should approach equilibrium for longer-period (e.g., synoptic scale) variations. No comparisons with field data were given.

A Coupled Energy Balance Climate–Sea Ice Model

Ledley [1988] developed a coupled heat balance model with land, ice, and atmosphere represented in 10° belts of latitude, meridionally averaged. The influence of leads was investigated by doubling, quadrupling, or eliminating the minimum fraction of open water in sea ice areas. Thermal feedback mechanisms of sensible and other heat fluxes in winter and of albedo and shortwave radiation in summer were discussed. The model shows a large change in sensible heat flux and air temperature in polar regions in winter in response to small changes in the minimum amount of open water. For example, eliminating leads reduced surface air temperature by up to 3°C in the north polar region in September and reduced air temperatures in all regions and seasons by lesser amounts. Increasing the lead fraction resulted in a global increase in simulated surface air temperature, with the largest increases in October in the northern hemisphere and in June in the southern hemisphere.

8. SUMMARY

Polynyas are localized features which occur intermittently in time. Their small size and restriction to certain portions (usually coastal) of the polar regions belie their significance within a regional oceanographic context. Significant evidence has been published in support of the hypothesis that quasi-continuous ice formation in latent heat polynyas on the shallow western Arctic shelves contribute, perhaps very significantly, to maintenance of the Arctic Ocean halocline. Our information concerning similar polynyas elsewhere on the shallow shelves which surround the Arctic Ocean is limited largely to that obtained from satellite imagery. Physical processes on these other shelves are probably similar to those in the western Arctic, so that they can be expected to play similarly significant roles relative to the larger-scale oceanography. Ice formation in Antarctic coastal polynyas can increase local salinities sufficiently to influence formation of Antarctic bottom water.

Latent heat polynyas typically occur in regions where the water is at the freezing point. Heat loss to the atmosphere leads to ice formation rather than to additional cooling. Sensible heat polynyas, however, present a different scenario. Since no ice (or very little ice) is formed, no salinity enhancement occurs. What does occur, however, is a very large sea-to-air heat flux which can locally condition the upper layer waters by cooling as opposed to brine enhancement. This appears to be a particularly important process for cooling, north flowing Atlantic water in the Whaler's Bay region north of Svalbard. It can also be expected to play a major role in deep ventilation of the Antarctic ocean.

The North Water polynya is a large area of relatively loose pack ice with diverging drift. The North Water Project led to a heat budget taking into account the distribution of pack ice and open water. While it was generally held that the North Water is maintained by southward drift of ice, this study concluded that advection of warmer water is also a major factor.

A wind-driven coastal polynya in the lee of the St. Lawrence Island varies in size according to wind speed and air temperature, and is a major source of ice in the Bering Sea. Our ability to predict occurrence and configuration of coastal latent heat polynyas such as this one through numerical models is quite good at this time, given adequate input data on meteorological conditions. Though it has not yet been done, these models can presumably be extended to more complex situations such as the North Water.

Prediction of the occurrence and configuration of sensible heat polynyas requires, in many cases, a better knowledge of the underlying oceanography than we have at present. For instance, we do not know why these features may occur in some years and be absent in others. The best example of this would be the Weddell Sea polynya, which has not reappeared since the late 1970s even though we believe we understand the mechanism of its formation. The contribution of oceanic heat to the maintenance of the North Water and Northeast Water polynyas remains uncertain. Ongoing research on these and related problems in the polar regions promises to improve our understanding of such processes and of the extent to which they influence the world ocean.

Leads are transient linear openings found throughout all areas of pack ice although their frequency of occurrence varies widely by region and sea. Leads do not occupy specific locations, as do polynyas, but may occur preferentially in specific regions. Like polynyas, leads lose heat in winter at a rate of several hundred watts per square meter, and the net heat loss from a small percentage of open water and thin ice exceeds that from the surrounding ice pack. This heat loss is accompanied by rapid freezing of ice and rejection of brine which is believed to sink in convection patterns to the base of the upper mixed layer or at times to penetrate to deeper layers.

Several experiments have studied atmospheric and oceanographic processes associated with leads and polynyas. The *Miyake* [1961] experiment established techniques and marked an early application of modern boundary-layer theory. The AIDJEX Lead Experiment of 1974, planned as a

multidisciplinary study of Arctic leads, was particularly successful in determining turbulent fluxes of sensible heat over natural and artificial leads. The oceanographic and atmospheric radiation results were fragmentary. A difficulty was that deployment time of the experiment was comparable to the lifetime of leads and that a limited deployment radius did not always encompass a suitable lead. The Dundas Polynya Experiment was successful in determining sensible heat flux and verifying a parameterization scheme which was used to calculate a surface heat budget for the duration of the experiment. Current and ocean temperature measurements in the strong tidal current regime of the area showed that advection of water up to 0.2°C above the freezing point was sufficient to maintain a group of polynyas in the area.

Atmospheric boundary layer fluxes at leads and small polynyas are basically the same, and progress has been made in understanding and modeling these atmospheric processes. The structure and extent of oceanic convection beneath leads remains an area of speculation. Radiative properties of the ice fog which typically overlies leads and polynyas also remain an area of conjecture.

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